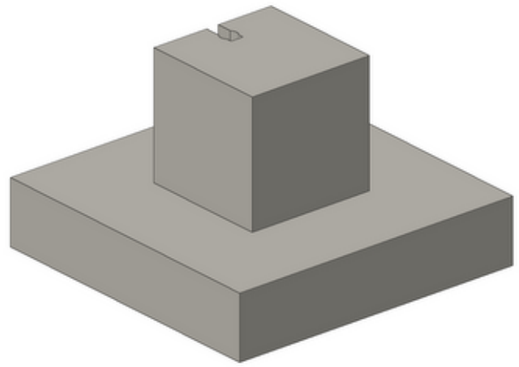
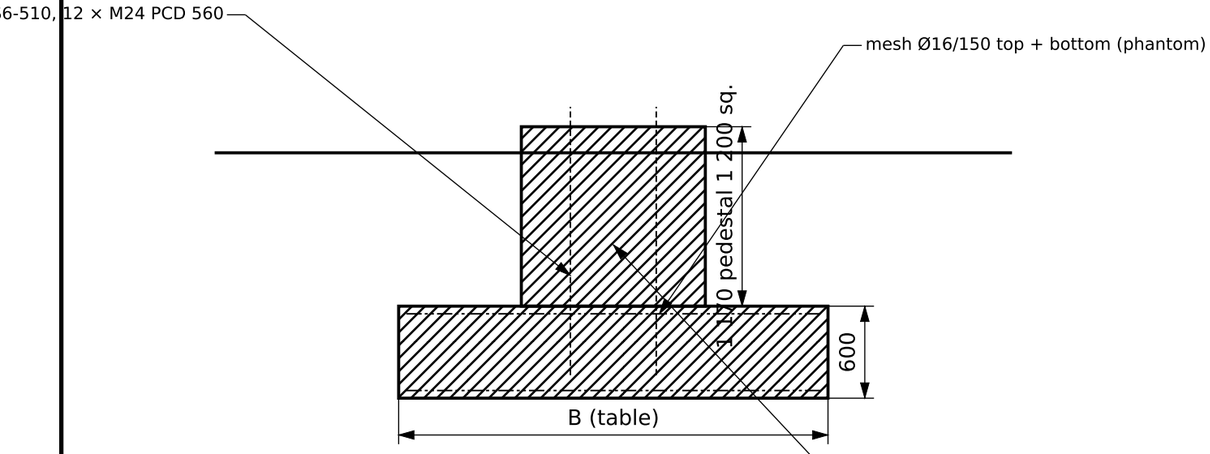
AXONOMETRIC VIEW (not to scale)

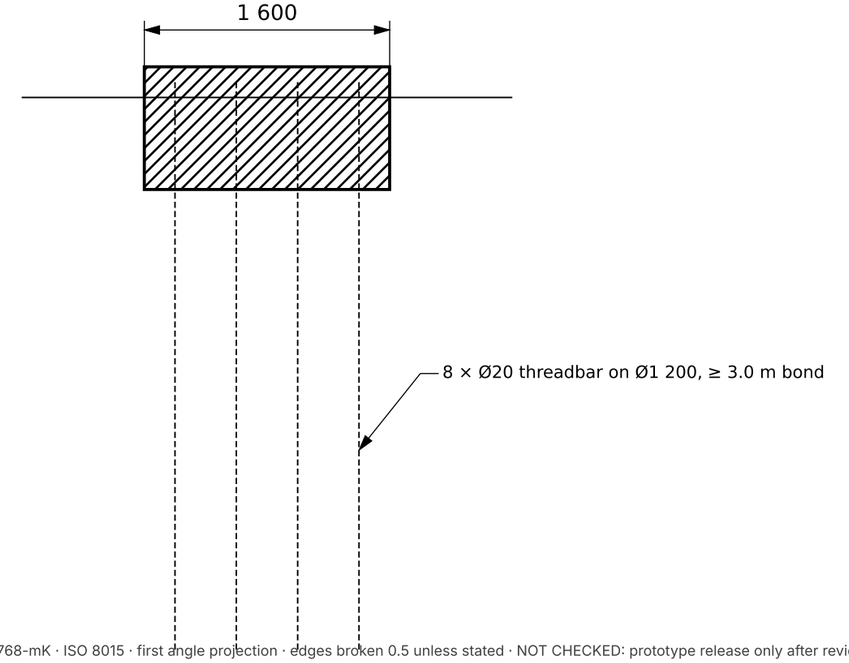


SPREAD FOOTING, SECTION



Ground	B m	Concrete m³	Bearing kPa	q_rd kPa
firm soil (dense gravel, moraine)	2.8	5.5	178.6	300.0
soft volcanic soil (loose ash, pumice sand)	3.1	6.6	99.0	100.0
frost-susceptible silt (thawed)	2.9	5.9	135.5	150.0
rock (anchored plinth)	1.6	2.0	–	anchors 92.1 kN

ROCK PLINTH (lava, basalt)



NOTES

1. Base 1.6 m below ground (frost depth); pedestal 1.2 × 1.2 m to +0.17, grout 30 to +0.20 under the base flange. Design base moment 238.6 kNm (earthquake 0.5 g + operation governs), parked V_e50 175.3 kNm.
2. Characteristic eccentricity ≤ B/4, operation ≤ B/6, bearing on the effective area ≤ q_rd (EN 1997 approach 2 style), as doc 09.
3. Concrete C30/37 XF4 air-entrained, cover 50; mesh Ø16/150 both ways top and bottom, pedestal 12 × Ø16 verticals + Ø10/150 links; earthing ring 50 mm² Cu round the footing, 2 rods 3 m.
4. Soil class to be confirmed by a site inspection; soft volcanic ash needs a plate-load test. Rock: drill Ø76, grout, proof-load each anchor to 1.25 × design.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FOUNDATION · SPREAD FOOTING AND ROCK PLINTH (8 m)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-500	B · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C30/37 XF4 air-entrained; B500C rebar	5.5–6.6 m³ (soil), 2.0 m³ (rock)	Claude / not checked